

Siemens LOGO! PLC — Ladder Logic Specification

 **Print-ready PDF:** docs/pdf/plc-ladder-logic.pdf

This document specifies all function blocks for the LOGO! Soft Comfort program. Program the LOGO! from this spec before connecting the rig to the RPi HMI.

LOGO! Model

LOGO! 12/24RCE (6ED1052-1MD08-0BA1) - Supply: 12/24V DC - Inputs: 8 digital (24V DC), 4 analog - Outputs: 4 relay (250V AC, 10A each) - Expansion: up to 24 I/O via expansion modules - Ethernet: built-in for LOGO! Web Server + Modbus TCP (port 502) - Software: LOGO! Soft Comfort v8.x (included on CD)

I/O Assignment

Terminal	Type	Signal name	Description	Switch type
I1	Input	ESTOP	E-stop mushroom button	NC
I2	Input	DOOR_INTERLOCK	Chamber door reed switch	NC
I3	Input	OVERTEMP_HW	120°C bimetallic thermostat cutout	NO
I4	Input	MANUAL_START	Green momentary start button	NO
I5	Input	MANUAL_STOP	Red momentary stop button	NC
I6–I8	Input	SPARE	Reserved for future expansion	—
Q1	Output	HEATER_RELAY	SSR control — heater circuit	Relay NO
Q2	Output	SERVO_POWER	24V→5V BEC power enable for servos	Relay NO
Q3	Output	DUT_POWER	DUT power enable relay	Relay NO
Q4	Output	ALARM_BEACON	External 24V alarm beacon	Relay NO

Modbus TCP Register Map

Enable Modbus server in LOGO! Soft Comfort: **Tools** → **Ethernet** → **Modbus server** → **Enable**.

Software enable inputs (written by RPi — M marker coils)

The RPi writes these marker flags as "software requests". The ladder ANDs each one with the hardware safety gate before driving the physical relay. This ensures the E-stop, door interlock, and overtemp cutout always take precedence over software.

Modbus address	Access	LOGO! variable	Description
Coil 2	R/W	M3	SW_HEATER — RPi software heater request → gates Q1
Coil 3	R/W	M4	SW_SERVO — RPi software servo-power request → gates Q2
Coil 4	R/W	M5	SW_DUT — RPi software DUT-power request → gates Q3

Physical output read-back (Q coils — read only from RPi)

Modbus address	Access	LOGO! variable	Description
Coil 8192	R	Q1	Heater relay actual state (read-back)
Coil 8193	R	Q2	Servo power relay actual state
Coil 8194	R	Q3	DUT relay actual state
Coil 8195	R	Q4	Alarm beacon actual state (ladder-only)

Physical input monitoring (discrete inputs — read only)

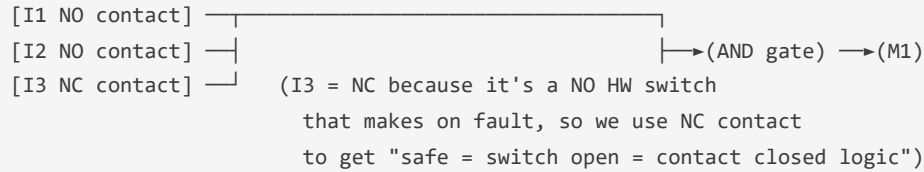
Modbus address	Access	LOGO! variable	Description
Input 8192	R	I1	E-stop physical state (1=safe)
Input 8193	R	I2	Door interlock (1=closed)
Input 8194	R	I3	HW overtemp (1=tripped)
Input 8195	R	I4	Manual start button (1=pressed)
Input 8196	R	I5	Manual stop button (1=not-pressed=safe)

Function Block Program

Network 1 — Safety gate (internal flag M1)

M1 is HIGH only when ALL physical safety conditions are met. M1 feeds into all output rungs.

FBD Network 1:



Note: I1 and I2 are NC switches → when safe, they provide 24V → LOGO! reads HIGH → use NO contact in ladder. I3 is NO switch → when tripped, provides 24V → use NC contact so M1 goes LOW when I3 is HIGH (tripped).

Network 2 — Run flag SR latch (M2)

SET input: [I4 NO] (manual start button)
RESET input: [I5 NC inverted] OR [M1 inverted] (stop button OR safety gate lost)
Output: →(M2 = "run permitted")

Network 3 — Q1 Heater relay

[M1 NO] —[M2 NO] —[M3 NO (SW_HEATER)]→(Q1)

Heater energises only when: safety OK AND run latched AND RPi software requests it (M3).

Network 4 — Q2 Servo power relay

[M1 NO] —[M4 NO (SW_SERVO)]→(Q2)

Servo power requires only the safety gate and software request (M4). No run latch needed — servos may be positioned during idle or cool-down.

Network 5 — Q3 DUT power relay

[M1 NO] —[M2 NO] —[M5 NO (SW_DUT)]→(Q3)

DUT power requires all three conditions like the heater. M5 is the software request written by the RPi.

Network 6 — Q4 Alarm beacon

[I1 NC inverted] —OR—[I2 NC inverted] —OR—[I3 NO]→(Q4)

Alarm fires immediately on any safety fault, independent of all software state. Cannot be silenced by writing M markers — only clearing the hardware fault extinguishes it. - NOT(I1) = alarm when E-stop tripped (I1 LOW → inverted → HIGH → alarm) - NOT(I2) = alarm when door open (I2 LOW → inverted → HIGH → alarm) - I3 = alarm when HW overtemp active (I3 HIGH → alarm)

LOGO! Soft Comfort Setup Procedure

1. Install LOGO! Soft Comfort from the CD included with the PLC.
 2. Open Soft Comfort → **File** → **New** → select **LOGO! 8 / 12/24RC**.
 3. Set the PC's network adapter to the same subnet as the LOGO! (default: 192.168.0.x).
 4. Connect USB programming cable (or Ethernet if your LOGO! model supports it).
 5. **Tools** → **Ethernet settings** → **IP address**: set to `192.168.1.100` (or match `rig_config.json`).
 6. **Tools** → **Ethernet** → **Modbus server**: enable, port 502.
 7. Enter the FBD program as specified in Networks 1–6 above.
 8. **File** → **Transfer to LOGO!** — wait for transfer confirmation.
 9. Set LOGO! to **RUN mode** via the front-panel button or Soft Comfort.
 10. Verify: press Manual Start (I4) → M2 should latch. Check with Soft Comfort online monitor.
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Verification Checklist

Before connecting the heater or DUT:

- [] LOGO! powered up, showing RUN on display
 - [] Modbus TCP responding on port 502 (test with `logo_driver.py ping()`)
 - [] I1 reads HIGH (E-stop button released, NC circuit closed)
 - [] I2 reads HIGH (door closed)
 - [] I3 reads LOW (HW thermostat under 120°C, NO contact open)
 - [] Pressing Manual Start → M2 latches → Q1/Q3 coils can be enabled via software
 - [] Pressing E-stop → I1 goes LOW → M1 goes LOW → Q1, Q3 de-energise immediately
 - [] Q4 alarm fires when E-stop pressed
 - [] Modbus write to Q1 coil with safety gate open → Q1 relay clicks
 - [] Modbus write to Q1 coil with E-stop pressed → Q1 does NOT energise
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